Geography Ch-4 (Advanced) — Denudation, Landslides & Groundwater in India

Advanced / India-applied | GS-I & GS-III (Disaster) | Source: D.R. Khullar (India: A Comprehensive Geography)

Complements Must-Do Ch-4 (Weathering/Mass movement/Groundwater) with India cases

Extra CA: Joshimath land subsidence (distinct from Must-Do's Dharali/Katra)

Facts from official sources | Inferences clearly marked | No fabricated data

■ HIGH UPSC RELEVANCE

HIGH

ADVANCED — India's Soil Erosion, Landslides & Groundwater Stress (D.R. Khullar)

GS-I / GS-III · India Geography → Denudation & Hazards (Source: D.R. Khullar, India: A Comprehensive Geography)

■ NEWS TRIGGER

Joshimath (Uttarakhand) land-subsidence crisis — ISRO/NRSC satellite data showed the town sinking, cracking hundreds of buildings; it exemplifies mass movement on weathered, over-loaded Himalayan slopes (distinct from the Dharali/Katra landslides in the Must-Do note).

■ INTRO & ORIGIN

Khullar treats weathering, soil erosion, landslides and groundwater as India's key denudation-linked hazards — critical for the fragile Himalaya and the intensively farmed plains.

Rock weathering feeds soil formation but, once vegetation is stripped and slopes over-steepened, the same loose material is rapidly eroded or fails as landslides — accelerated by monsoon rain and human activity.

■ KEY DATA

Process / type	Mechanism	Indian example / note
Sheet erosion	Uniform removal of topsoil by rain-wash	Widespread on cleared slopes
Rill & gully erosion	Channels deepen into gullies	Ravines of the Chambal (badlands)
Wind erosion	Creep, saltation, suspension	Thar/Rajasthan arid margins
Landslides	Slope failure on weathered debris	Himalaya, Western Ghats (monsoon-triggered)
Liquefaction / subsidence	Saturated ground loses strength	Joshimath sinking; Bhuj liquefaction

■ STATIC THEORY

• **Soil erosion** in India ranges from gentle **sheet** erosion to destructive **rill and gully** erosion, whose extreme form is the **ravine/badland** topography of the Chambal.

• **Wind erosion** (surface creep, saltation, suspension) dominates the arid Rajasthan margins; **glacial erosion** acts in the higher Himalaya.

• **Landslides** concentrate in the tectonically young, weathered, over-steepened Himalaya and the Western Ghats, and are triggered by intense monsoon rain, road-cutting and deforestation.

• **Land subsidence** (e.g. Joshimath) occurs where weathered, water-saturated debris is over-loaded by unplanned construction — a slow mass-movement hazard.

• **Groundwater** in India faces both over-extraction (falling water-tables) and **contamination** — waste, fluids via wells/sinkholes infiltrating aquifers.

■ MUST-KNOW FACTS

1. Erosion sequence: sheet → rill → gully → ravine (Chambal badlands).

2. Wind erosion transport: surface creep, saltation, suspension (Rajasthan).

■■ UPSC TRAPS

✗ Gully erosion and sheet erosion are equally destructive.

✓ Sheet erosion is gradual/uniform; gully erosion is far more destructive and ends in ravine/badland topography.

3. Landslide hot-spots: Himalaya & Western Ghats (monsoon-triggered).	✗ Land subsidence and landslides are unrelated to weathering.
4. Joshimath = land subsidence on weathered, over-loaded slope (ISRO/NRSC).	✓ Both act on weathered, loose regolith — subsidence (Joshimath) and slides are mass-movement outcomes of weathering + loading/water.
5. Groundwater threats: over-extraction + aquifer contamination via sinkholes/wells.	✗ Groundwater problems in India are only about quantity.
6. Liquefaction of saturated sands worsens quake damage (Bhuj 2001).	✓ Both QUANTITY (over-extraction) and QUALITY (aquifer contamination) are critical concerns.

■ MAINS ANGLE

Analyse how weathering + mass movement + human loading produce India's slope hazards (Joshimath, Himalayan/Ghats landslides) and argue for hazard zonation, carrying-capacity limits and slope management.

■ STUDY LINK

Geography → India → Denudation, Hazards & Groundwater (Khullar, Natural Hazards & Environment)

End of Register Notes — UPSC Agent / Copilot CLI